

FILE : CH9PP1

```
str = input("Enter the string: ")
ch = input("Enter the character to count: ");
c = str.count(ch)
print(ch, "occurs", c, "times")
```

FILE : CH9PP2

```
str = input("Enter the string: ")
newStr = ""
for ch in str :
    lch = ch.lower()
    if lch in 'aeiou':
        newStr += '*'
    else :
        newStr += ch
print(newStr)
```

FILE : CH9PP3

```
str = input("Enter the string: ")
newStr = ""
for ch in str :
    newStr = ch + newStr
print(newStr)
```

FILE : CH9PP3A

```
str = input("Enter the string: ")
for ch in range(len(str)-1,-1,-1) :
    print(str[ch],end='')

```

FILE : CH9PP4

```
phNo = input("Enter the phone number: ")
length = len(phNo)
if length == 12 \
    and phNo[3] == "-" \
    and phNo[7] == "-" \
    and phNo[:3].isdigit() \
    and phNo[4:7].isdigit() \
    and phNo[8:].isdigit() :
    print("Valid Phone Number")
else :
    print("Invalid Phone Number")
```

FILE : CH9PP5

```
str = input("Enter the string: "); sum = 0; digitStr = ''
for ch in str :
    if ch.isdigit() :
        digitStr += ch
        sum += int(ch)
if not digitStr :
    print(str, "has no digits")
else :
    print(str, "has the digits", digitStr, "which sum to", sum)
```

FILE : CH9PP6

```
str = input("Enter a few sentences: "); length = len(str);
spaceCount = 0
alnumCount = 0
for ch in str :
    if ch.isspace() :
        spaceCount += 1
    elif ch.isalnum() :
        alnumCount += 1
alnumPercent = alnumCount / length * 100
print("Original Sentences:")
print(str)
print("Number of words =", (spaceCount + 1))
print("Number of characters =", (length))
print("Alphanumeric Percentage =", alnumPercent)
```

FILE : CH9PP7

```
while True :
    str = input("Please enter a sentence, or 'q' to quit : ")
    newStr = ""
    if str.lower() == "q" :
        break
    for ch in str :
        if ch.islower() :
            newStr += ch.upper()
        elif ch.isupper() :
            newStr += ch.lower()
        else :
            newStr += ch
    print(newStr)
```

FILE : CH9PP8

```
num = int(input("Enter an integer: "))
str = input("Enter the string: ")
digitsStr = ''
digitsNum = 0;
for ch in str :
    if ch.isdigit() :
        digitsStr += ch
if digitsStr :
    digitsNum = int(digitsStr)
print(num, "+", digitsNum, "=", (num + digitsNum))
```

FILE : CH9PP9

```
str1 = input("Enter first string: ")
str2 = input("Enter second string: ")
small = str1; large = str2
if len(str1) > len(str2) :
    large = str1
    small = str2
print(small)
lenLarge = len(large)
for i in range(lenLarge // 2) :
    print(' ' * i, large[i], ' ' * (lenLarge - 2 * i),
large[lenLarge - i - 1], sep='')
```

FILE : CH9PP10

```
# printing of roman no
n = int(input("Enter the number: "))
num = (1000, 900, 500, 400, 100, 90, 50, 40, 10, 9, 5, 4, 1)
rom = ('M', 'CM', 'D', 'CD', 'C', 'XC', 'L', 'XL', 'X', 'IX', 'V', 'IV', 'I')
result = ''
for i in range(len(num)) :
    count = int(n / num[i])
    result += str(rom[i] * count)
    n -= num[i] * count
print(result)
```

FILE : CH9PP10A

```
# printing of roman no
n=int(input("Enter the number: "))
num=(1000,900,500,400,100,90,50,40,10,9,5,4,1)
rom=('M','CM','D','CD','C','XC','L','XL','X','IX','V','IV','I')
result=''
for i in range(len(num)) :
    count = int(n / num[i])
    result += str(rom[i] * count)
    n -= num[i] * count
print(result.lower())
```

FILE : CH9PP11

```
# to count the No. of words in a string(sentence)
str = input("Enter a string: ")
count = 0
for ch in str :
    if ch.isspace() :
        count += 1
print("No of words =", count)
```

FILE : CH9PP12

```
# to check equality of opening and closing () in a formula
str = input("Enter a formula: ")
count = 0
for ch in str :
    if ch == '(' :
        count += 1
    elif ch == ')' :
        count -= 1
if count == 0 :
    print("Formula has same number of opening and closing
parentheses")
else :
    print("Formula has unequal number of opening and closing
parentheses")

# rectify the Error
```

FILE : CH9PP13

```
str = input("Enter a string: ").lower()
count = 0
for ch in str :
    if ch in "aeiou":
        count += 1
print("Vowel Count =", count)
```

FILE : CH9PP13A

```
str = input("Enter a string: ").lower()
count = 0
for ch in str :
    if ch in "aeiou":
        count += 1
print("Vowel Count =", count)
```

FILE : CH9PP13B

```
# to display a string without the vowels
str = input("Enter a string: ").lower()
for ch in str :
    if ch not in "aeiou":
        print(ch,end="")
```

FILE : CH9PP13C

```
str = input("Enter a string: ").lower()
for ch in str :
    if ch in "aeiou":
        print(ch.upper(),end="")
    else:
        print(ch,end="")
```

FILE : CH9PP14

```
# to find the longest word in a string(sentence)
str = input("Enter a string: ")
words = str.split()
longWord = ''
for w in words :
    if len(w) > len(longWord) :
        longWord = w
print("Longest Word =", longWord)
```

FILE : CH9PP15

```
str = input("Enter a string: ")
words = str.split()
newStr = ""
for w in words :
    rw = ""
    for ch in w :
        rw = ch + rw
    newStr += rw + " "
print(newStr)
```

FILE : CH9PP15A

```
str = input("Enter a string: ")
words = str.split()
for w in words :
    rw = ""
    for ch in w :
        rw = ch + rw
    print(rw,end=" ")
```

FILE : CH9PP15B

```
str = input("Enter a string: ")+" "
rw=""
for w in str :
    if w==" " :
        print(rw,end=" ")
        rw = ""
    else:
        rw = w + rw
```

FILE : CH9PP15C

```
str = input("Enter a string: ")+" "
rw=""
for w in range(len(str)):
    if str[w]==" " :
        print(rw,end=" ")
        rw=""
    else:
        rw=str[w]+rw
```